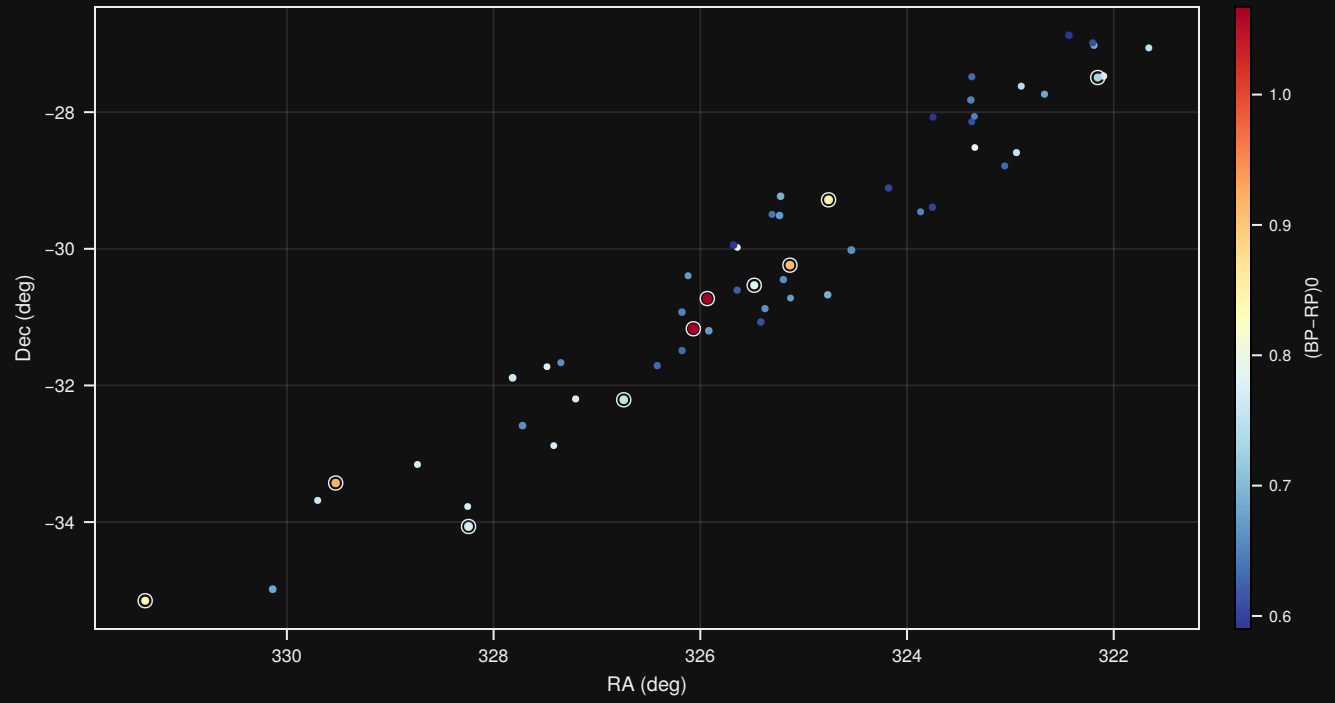
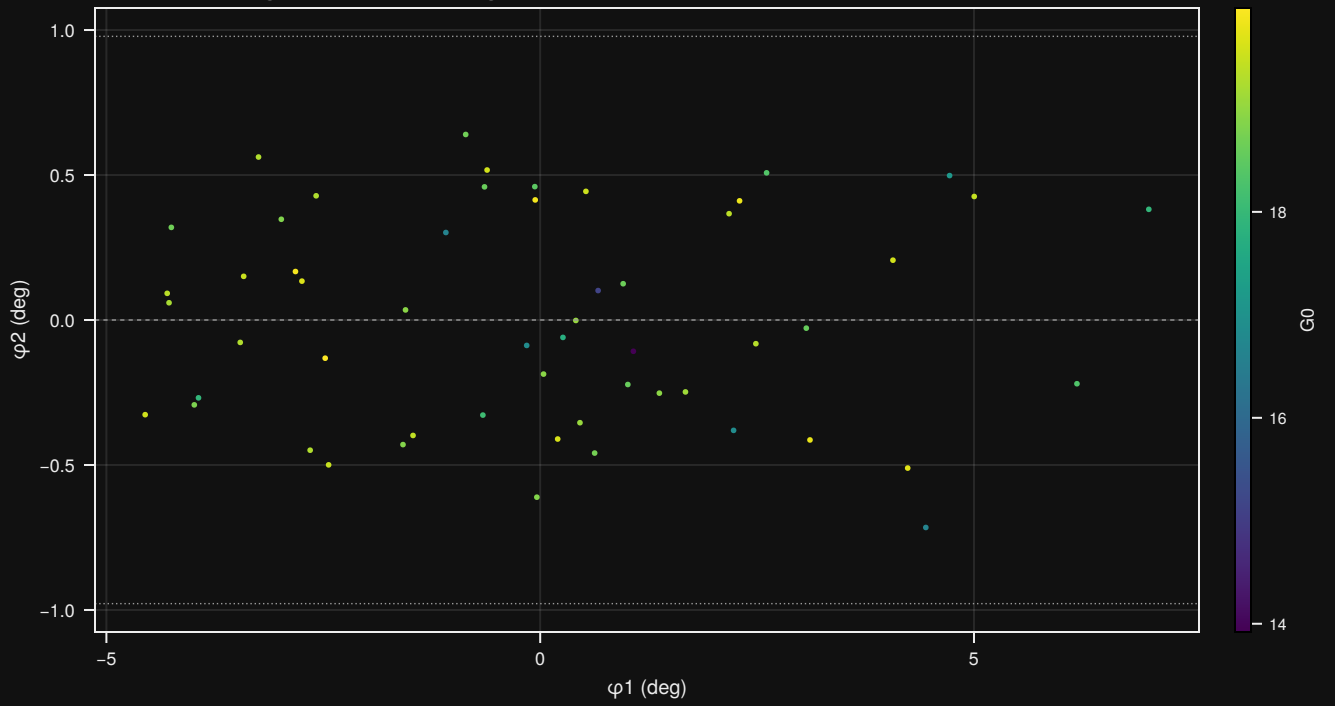
New-22

MEMBERSHIP	
Stars in catalog	55
with Gaia GSP-Phot	25 (45%)
with radial velocity	10
Stream id (sID)	78
GEOMETRY · great-circle fit	
Centroid RA, Dec	325.4, -30.4 deg
Centroid l, b	16.8, -48.4 deg
Pole RA, Dec	31.2, 35.0 deg
Length (φ1, 2–98%)	10.4 deg
Width σ(φ2), robust	0.49 deg
KINEMATICS	
pmRA* med ± σ	-3.50 ± 0.47 mas/yr
pmDE med ± σ	-0.93 ± 0.29 mas/yr
μφ1* med ± σ	-1.64 ± 0.28 mas/yr
μφ2 med ± σ	-3.26 ± 0.51 mas/yr
Parallax (cat.) med	0.097 ± 0.342 mas
Parallax (wtd, ZP-corr.)	0.158 ± 0.019 mas
Distance	d = 6.3 ± 0.8 kpc
Physical length	1.15 kpc
RV median ± σ(MAD)	15.3 ± 23.4 km/s
RV gradient dV/dφ1	1.20 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	13.9–20.0 (18.9)
(BP−RP)0 median	0.68
GSP-Phot [M/H] (n=25)	-1.33 ± 0.42
GSP-Phot Teff median	5720 K
RUWE > 1.4	0.0 %
PM clusters (largest)	3 (21 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	55 | 55
n_RV catalog | paper	10 | 10
<[Fe/H]> paper	-0.98
Stellar mass (paper)	580 Msun
Pole offset vs paper	0.2 deg
Remark	Velocities unclear

Sky map · catalog members (n = 55) · size ∝ brightness · ring = has RV

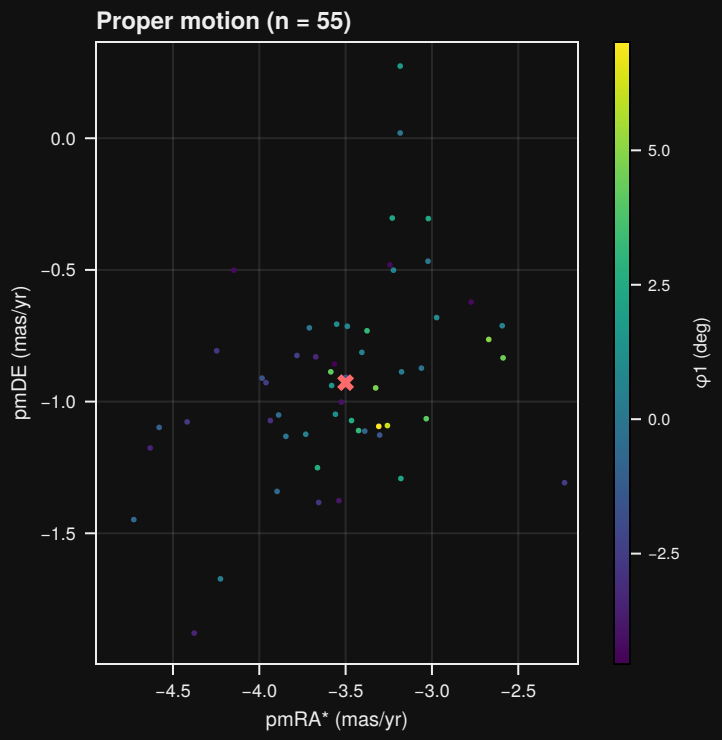


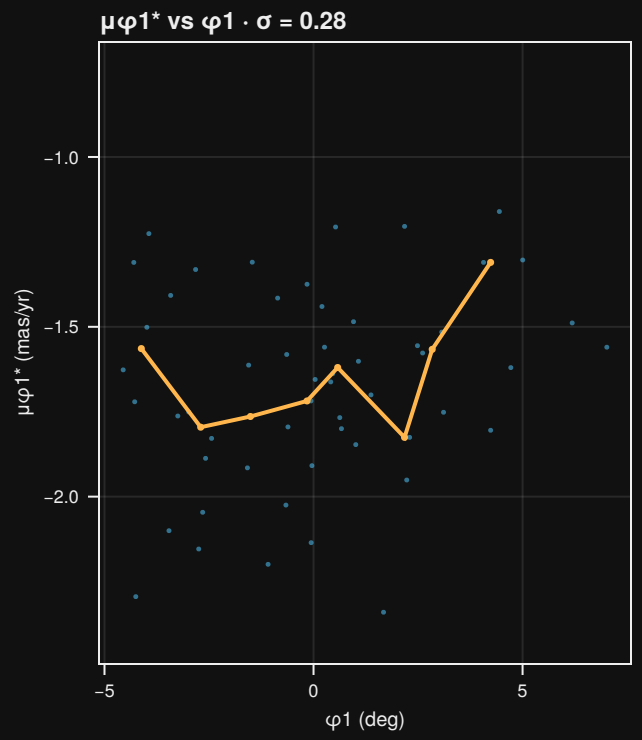
Stream frame (great-circle fit) · length 10.4° · width σ 0.49°

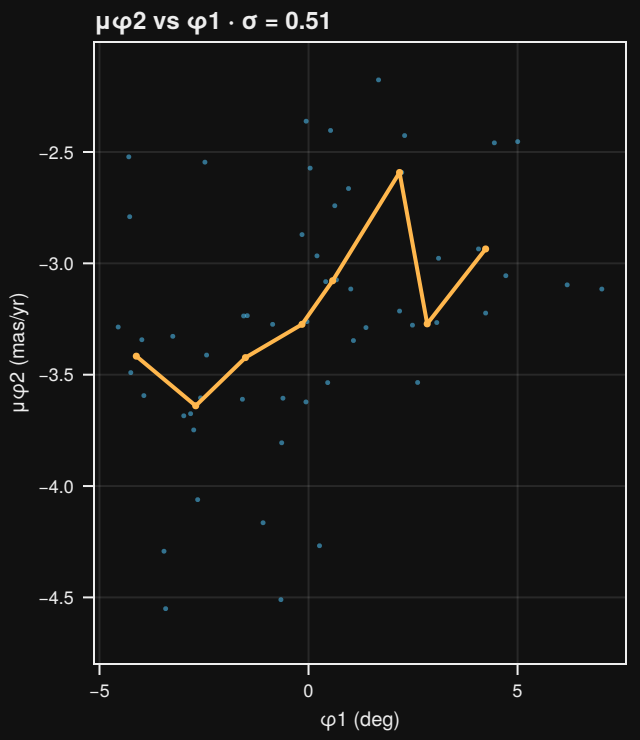


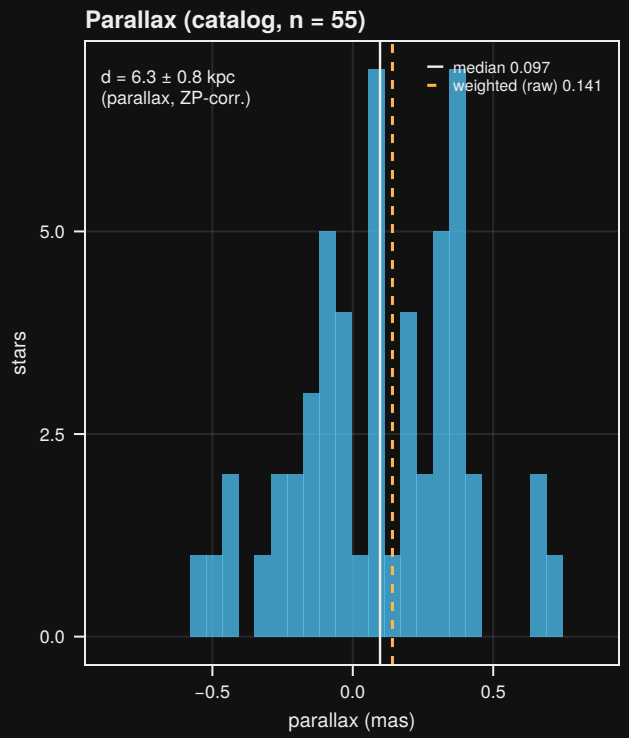
New-22

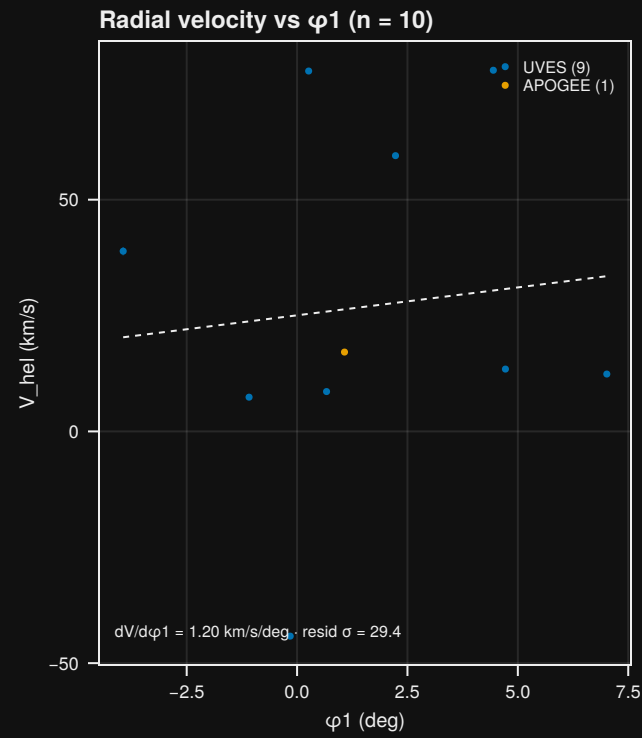
Kinematics · proper motion · parallax · radial velocity

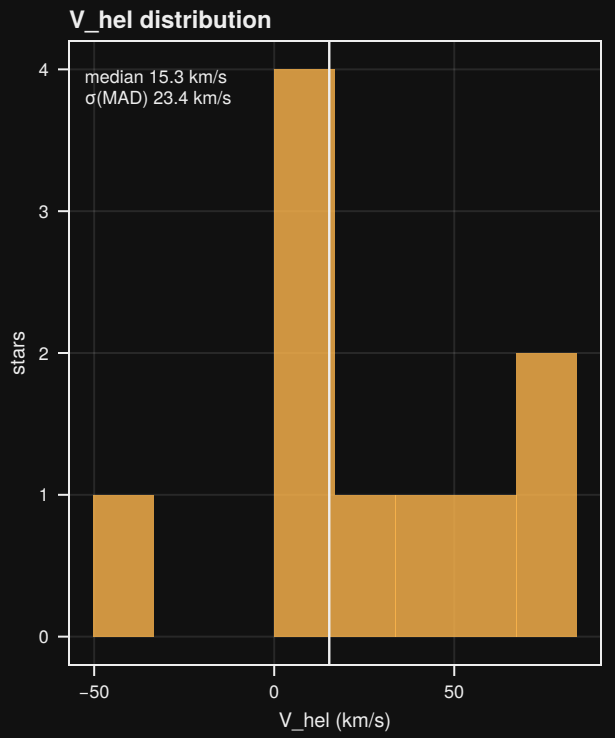






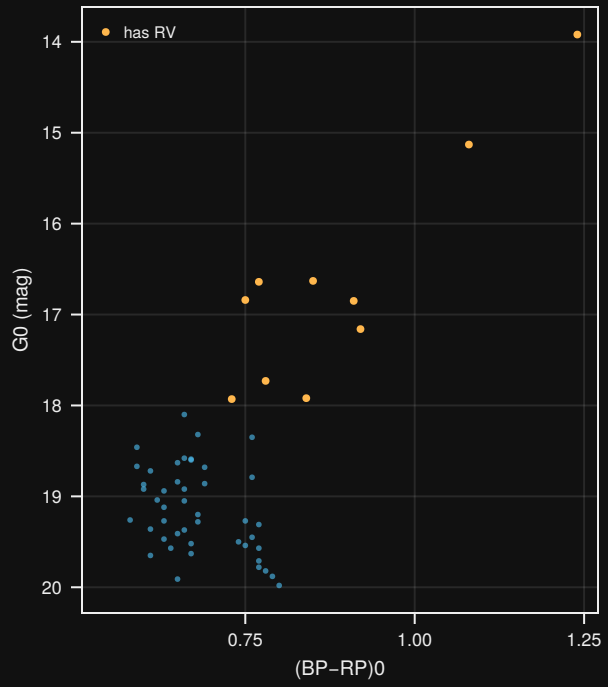




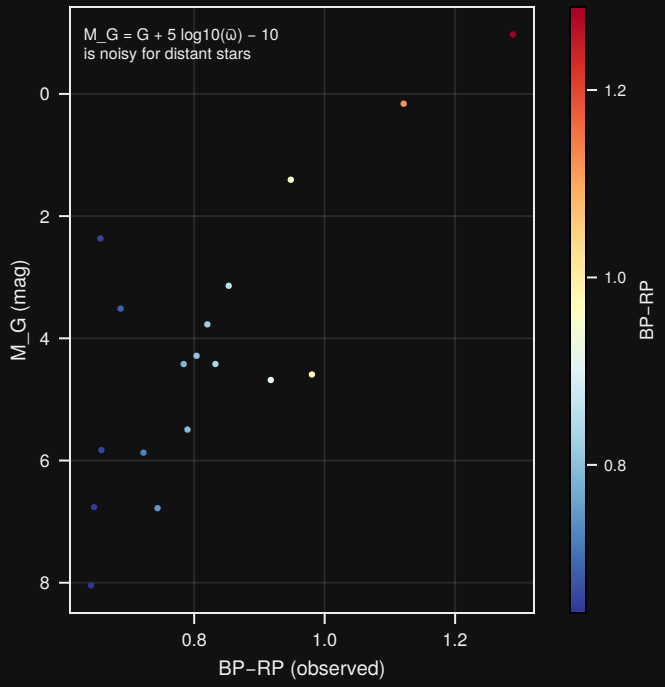


New-22

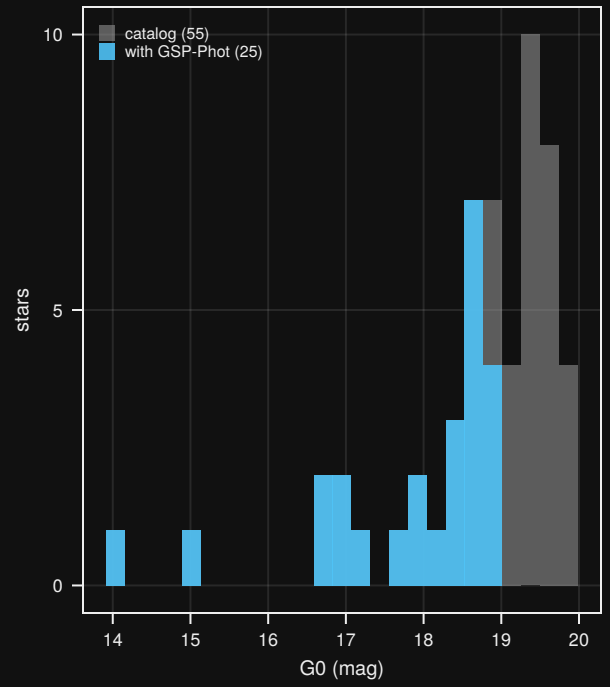
CMD · dereddened (n = 55)



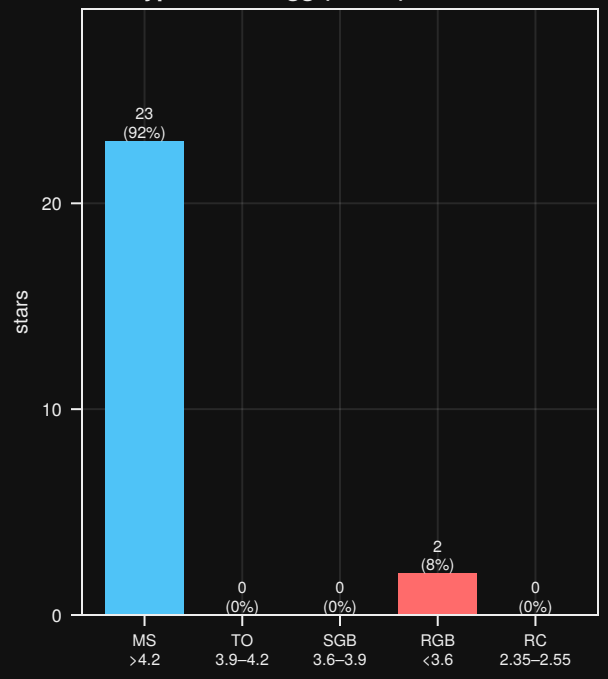
HR diagram · ϖ>0 (n = 18/25)



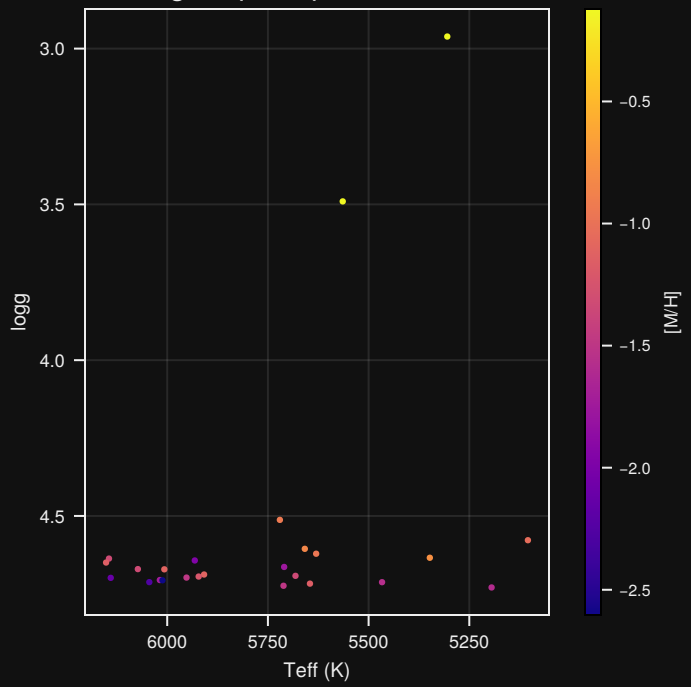
G0 & GSP-Phot coverage



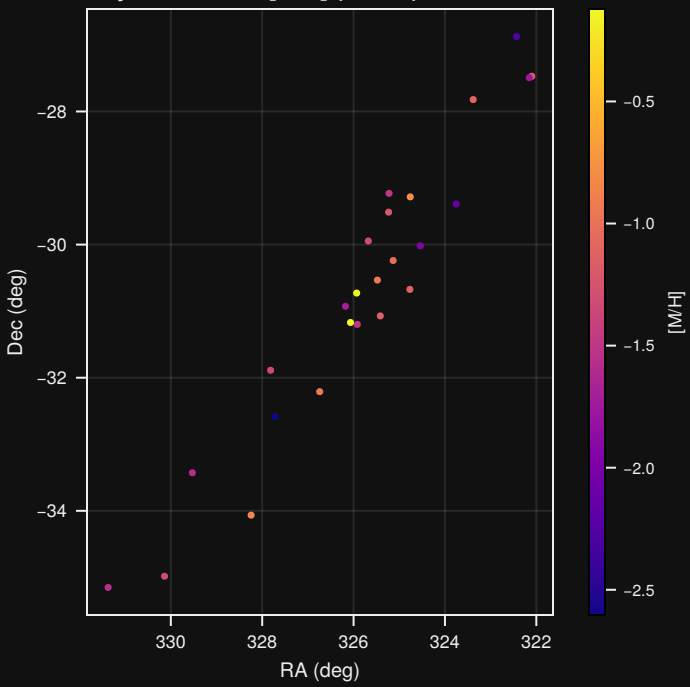
Star types from logg (n = 25)



Kiel diagram (n = 25)



Sky · GSP-Phot [M/H] (n = 25)

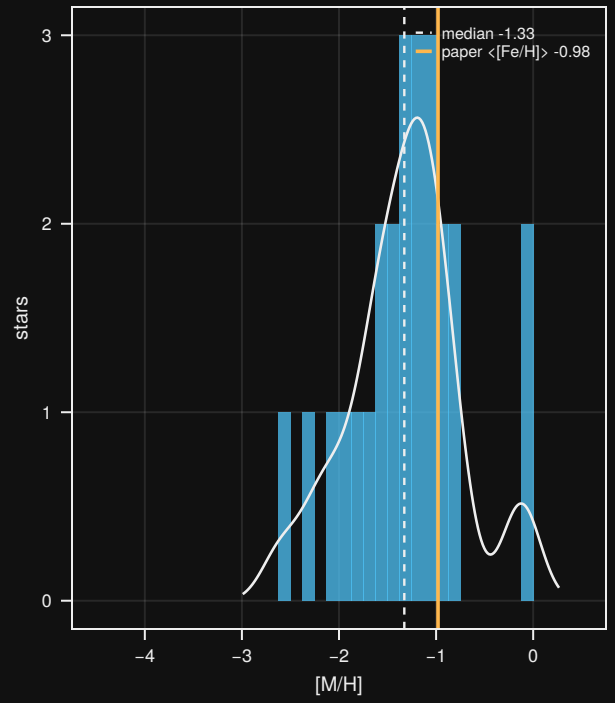


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

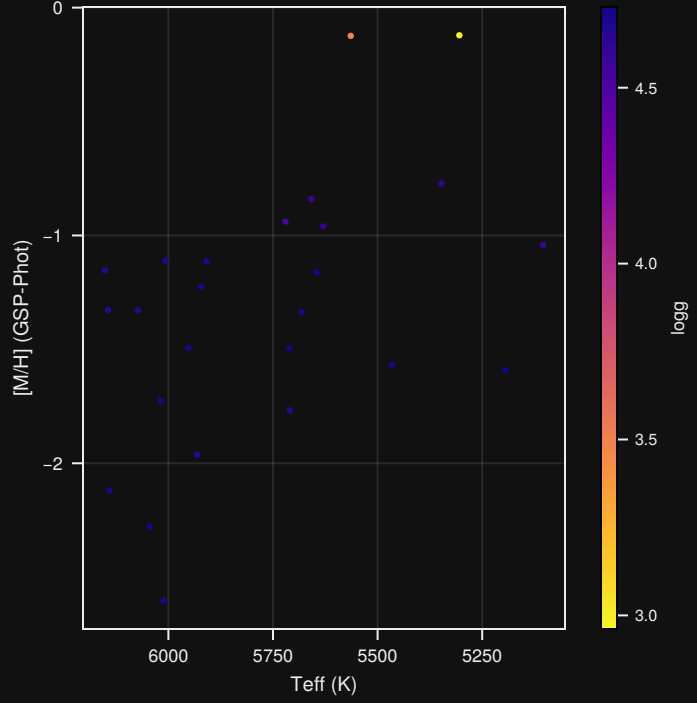
New-22

Metallicity · age · astrometric quality

[M/H] GSP-Phot (n = 25)



Teff vs [M/H] (n = 25)



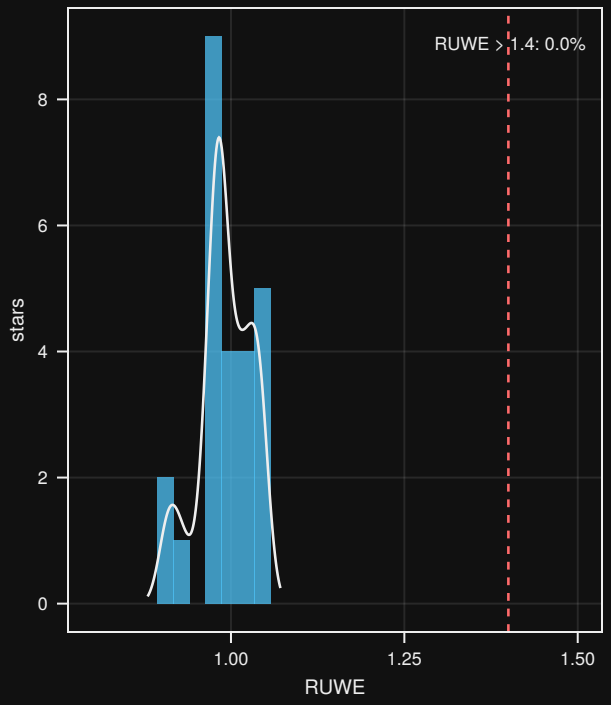
FLAME age (n = 2)

FLAME ages for only 2 star(s)
(FLAME needs good parallaxes)

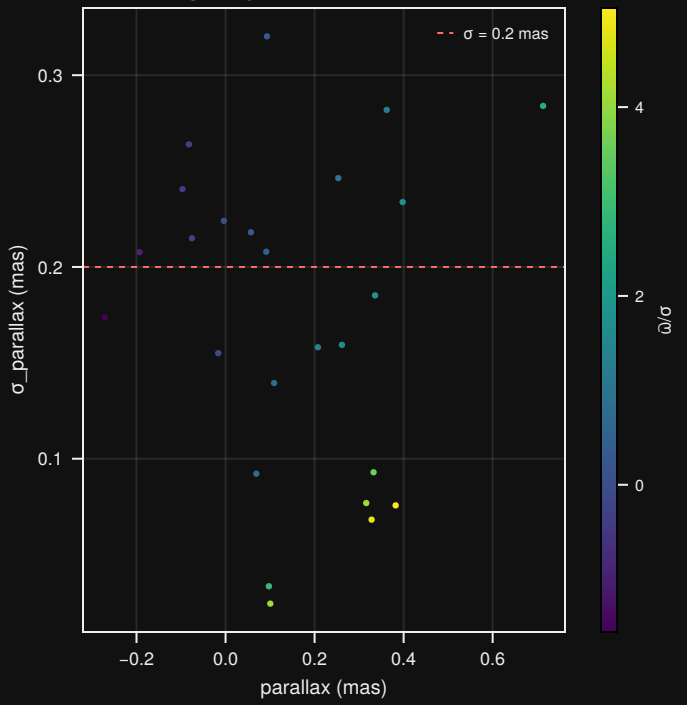
Age–metallicity (n = 2)

n < 3

RUWE (n = 25)

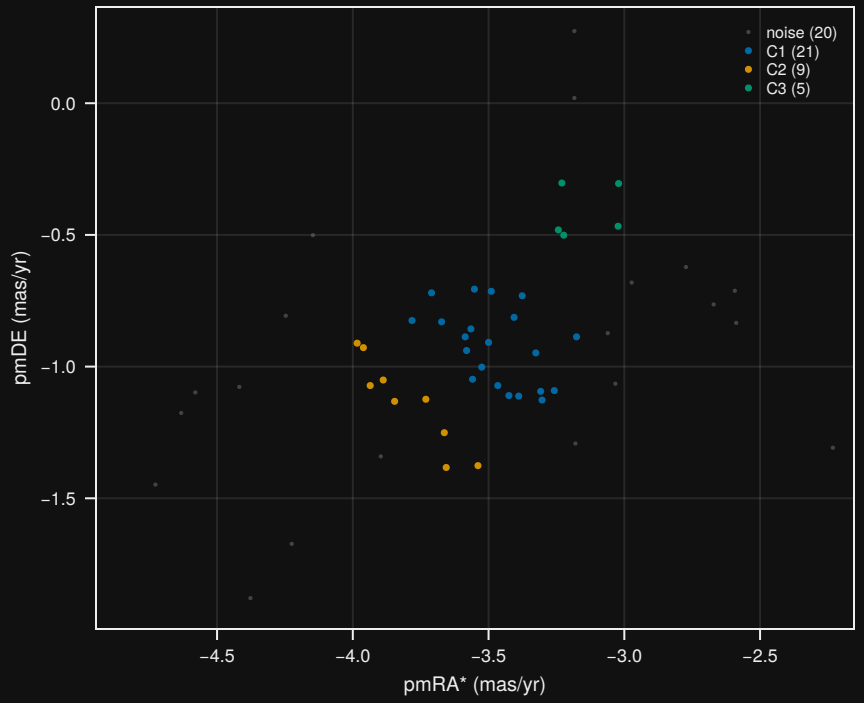


Parallax quality (n = 25)

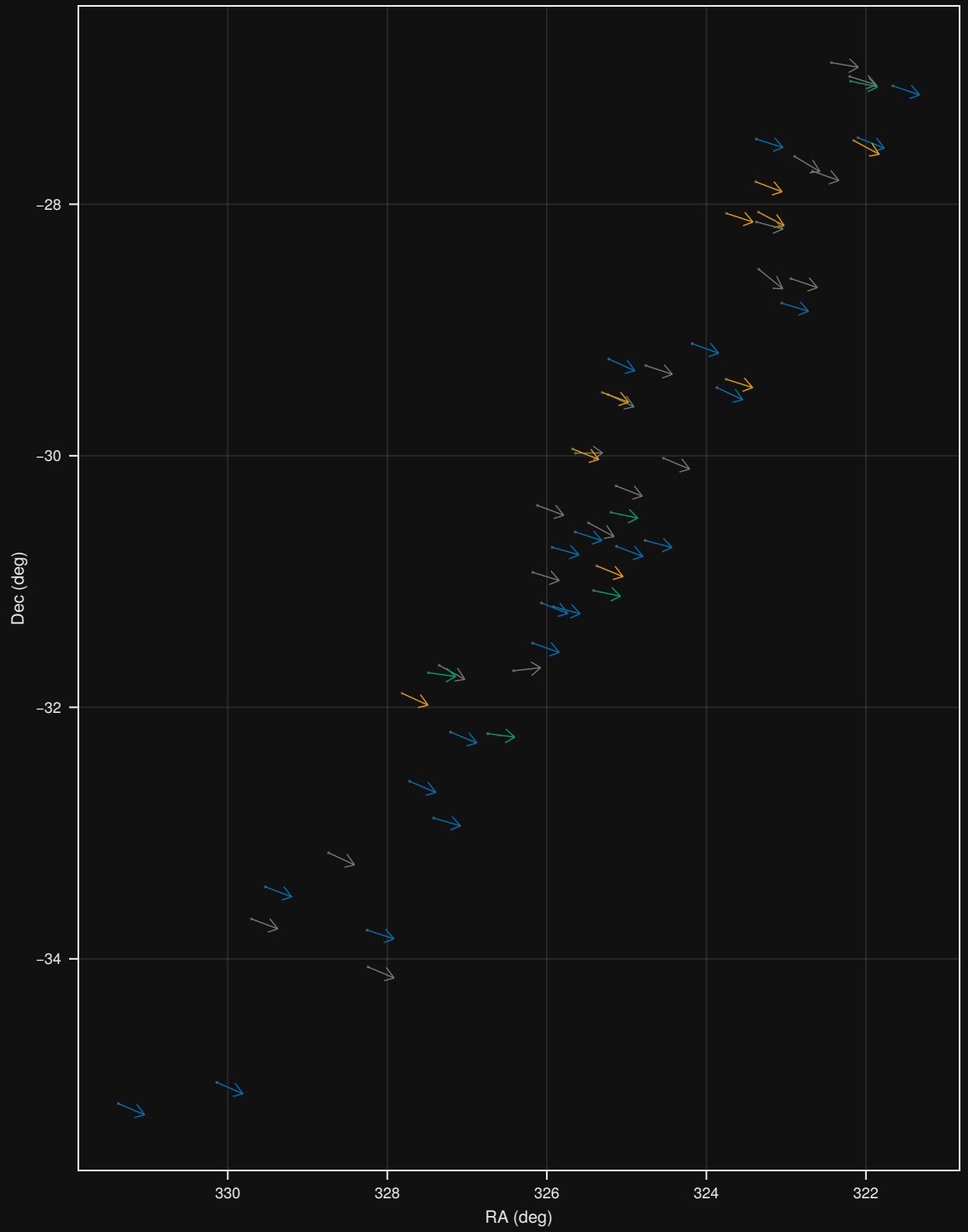


New-22

PM clustering · HDBSCAN · 3 cluster(s) · noise 36%



Proper-motion directions (arrows) coloured by cluster



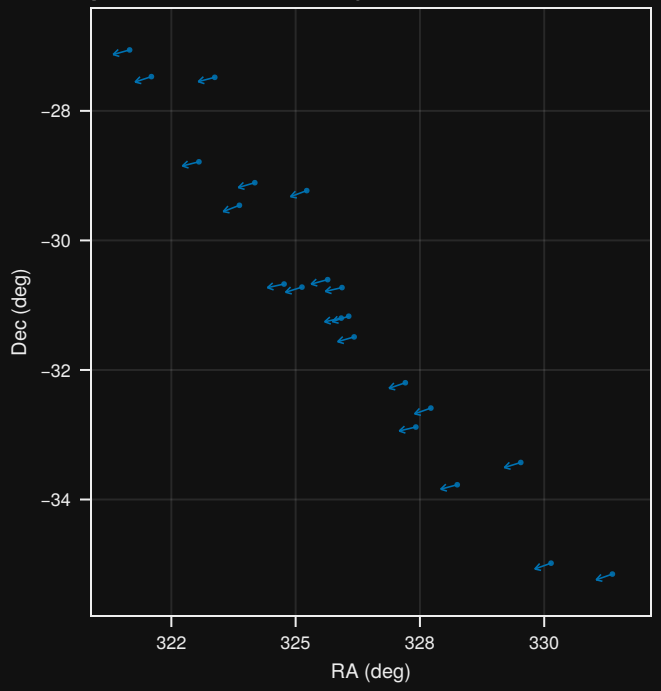
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	20	-3.18	-0.97	0.97	-1.13	4
C1	21	-3.49	-0.91	0.14	-1.41	4
C2	9	-3.85	-1.12	0.17	-1.34	1
C3	5	-3.22	-0.47	0.03	-1.05	1

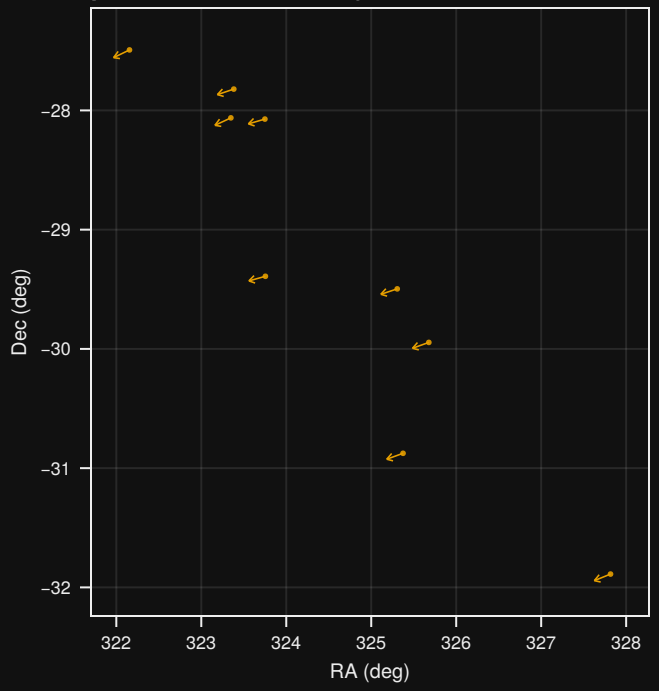
New-22

Per-cluster proper-motion vector maps

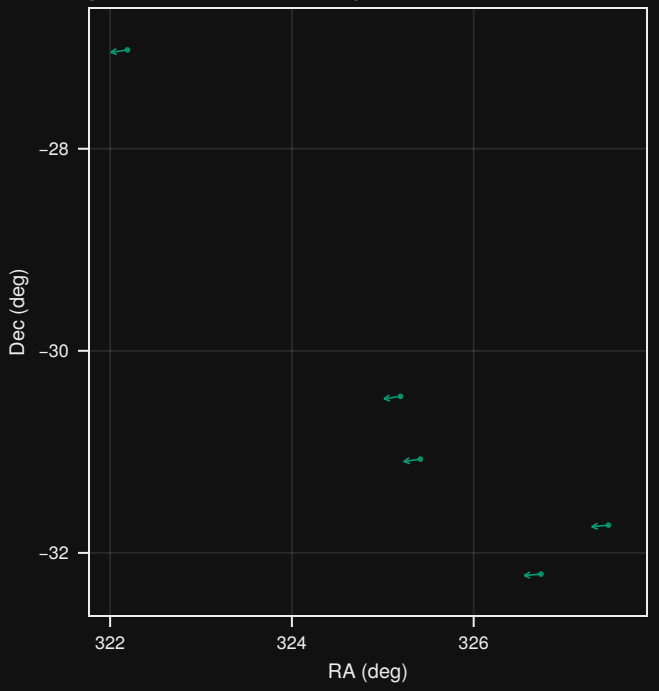
Cluster C1 · n = 21
pm = (-3.49, -0.91) mas/yr



Cluster C2 · n = 9
pm = (-3.85, -1.12) mas/yr



Cluster C3 · n = 5
pm = (-3.22, -0.47) mas/yr



Noise · n = 20

